

Homework 9

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Problem 1

Suppose that we roll a pair of fair dice until the sum of the numbers on the dice is seven. We want the expected number of times we roll the dice.

Solution. There are $6 \times 6 = 36$ possible outcomes for two dice. Six outcomes give a total of 7, so

$$p = \Pr(\text{sum} = 7) = \frac{6}{36} = \frac{1}{6}.$$

The number of rolls until the first 7 follows a geometric distribution with success probability $p = \frac{1}{6}$. The expected number of trials for a geometric random variable is $\frac{1}{p}$.

$$\mathbb{E}[X] = \frac{1}{p} = 6.$$

Therefore, the expected number of rolls is $\boxed{6}$.

Problem 2

Let A be an event. Define the indicator random variable of A as

$$1_A = \begin{cases} 1, & \text{if } A \text{ occurs,} \\ 0, & \text{otherwise.} \end{cases}$$

To show: $\mathbb{E}[1_A] = \Pr(A)$.

Solution. By the definition of expectation for a discrete random variable,

$$\mathbb{E}[1_A] = (1) \Pr(1_A = 1) + (0) \Pr(1_A = 0) = \Pr(1_A = 1).$$

Since $1_A = 1$ exactly when A occurs,

$$\Pr(1_A = 1) = \Pr(A).$$

Therefore,

$$\boxed{\mathbb{E}[1_A] = \Pr(A)}.$$

Problem 3

When m balls are distributed into n bins uniformly at random:

1. Probability that the first bin remains empty.

Each ball independently avoids the first bin with probability $1 - \frac{1}{n} = \frac{n-1}{n}$. Thus,

$$\Pr(\text{bin 1 empty}) = \left(\frac{n-1}{n}\right)^m.$$

2. Expected number of balls in the first bin.

Let X_i be an indicator variable for ball i landing in bin 1. Then $X = \sum_{i=1}^m X_i$, and $\mathbb{E}[X_i] = \frac{1}{n}$. By linearity of expectation,

$$\mathbb{E}[X] = \sum_{i=1}^m \mathbb{E}[X_i] = \frac{m}{n}.$$

3. Variance of the number of balls in the first bin.

Since X is the sum of m independent Bernoulli variables with $p = \frac{1}{n}$,

$$\text{Var}(X) = mp(1-p) = m \frac{1}{n} \left(1 - \frac{1}{n}\right) = \frac{m(n-1)}{n^2}.$$

4. Expected number of bins that remain empty.

Let J_j be an indicator variable that bin j is empty. Then

$$Y = \sum_{j=1}^n J_j.$$

Each bin has the same probability of being empty, so

$$\mathbb{E}[Y] = n \Pr(\text{a bin is empty}) = n \left(\frac{n-1}{n}\right)^m.$$

Problem 4

The sample space is the disjoint union $S = \bigsqcup_{j=1}^n S_j$. For any random variable X ,

$$\mathbb{E}[X] = \sum_{j=1}^n \mathbb{E}[X \mid S_j] \Pr(S_j).$$

Proof. Let 1_{S_j} be the indicator of S_j . Since the S_j 's form a partition,

$$X = \sum_{j=1}^n X 1_{S_j}.$$

Take expectations and use linearity:

$$\mathbb{E}[X] = \sum_{j=1}^n \mathbb{E}[X 1_{S_j}].$$

For each j , on S_j the variable 1_{S_j} equals 1, so

$$\mathbb{E}[X 1_{S_j}] = \mathbb{E}[\mathbb{E}[X 1_{S_j} \mid S_j]] = \mathbb{E}[1_{S_j} \mathbb{E}[X \mid S_j]] = \mathbb{E}[X \mid S_j] \Pr(S_j).$$

Summing over j gives the claim. □

Application. Let M be the event “male” and F the event “female,” with $\Pr(M) = 0.12$, $\Pr(F) = 0.88$. Given $\mathbb{E}[X \mid M] = 4200$ lb and $\mathbb{E}[X \mid F] = 1100$ lb,

$$\mathbb{E}[X] = \mathbb{E}[X \mid M] \Pr(M) + \mathbb{E}[X \mid F] \Pr(F) = 4200(0.12) + 1100(0.88) = 504 + 968 = \boxed{1472 \text{ lb}}.$$

Problem 5

Let $X \geq 0$ and $a > 0$. Show $\Pr(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$ (Markov).

Proof. Let $I = 1_{\{X \geq a\}}$. Then $X \geq aI$, so $I \leq X/a$. Take expectations:

$$\Pr(X \geq a) = \mathbb{E}[I] \leq \mathbb{E}\left[\frac{X}{a}\right] = \frac{\mathbb{E}[X]}{a}.$$

Problem 6

Let X be the number of cans filled in a day. Given $\mathbb{E}[X] = 10,000$ and $\text{Var}(X) = 1000$.

1. **Markov bound for** $\Pr(X > 11,000)$. Since $X \geq 0$, by Markov's inequality,

$$\Pr(X \geq 11,000) \leq \frac{\mathbb{E}[X]}{11,000} = \frac{10,000}{11,000} = \frac{10}{11} \approx 0.9091.$$

2. **Chebyshev lower bound for** $\Pr(9000 \leq X \leq 11,000)$. Here $\mu = 10,000$ and the half-width is $k = 1000$. Chebyshev gives

$$\Pr(|X - \mu| \geq k) \leq \frac{\text{Var}(X)}{k^2} = \frac{1000}{(1000)^2} = 0.001.$$

Hence

$$\Pr(9000 \leq X \leq 11,000) = \Pr(|X - \mu| \leq 1000) \geq 1 - 0.001 = 0.999.$$

Problem 7

Underwriting probability of a hole-in-one: $p = \frac{1}{12,500} = 0.00008$. Premium $P = \$180$. Prize (if a hole-in-one occurs) is \$1,000,000.

1. **Expected profit per contest.**

$$\mathbb{E}[\text{profit}] = P - p \cdot 1,000,000 = 180 - 80 = \boxed{\$100}.$$

2. **Kelly-optimal fraction.** Model liability that scales with bankroll W : choose fraction f so the policy limit is $L = fW$. Per \$1 of liability, the premium rate is $r = \frac{P}{1,000,000} = 0.00018$. The per-contest multiplicative return is

$$\begin{cases} 1 + fr, & \text{with prob } q = 1 - p, \\ 1 - f(1 - r), & \text{with prob } p. \end{cases}$$

Maximize $q \log(1 + fr) + p \log(1 - f(1 - r))$. Setting the derivative to zero gives

$$f^* = \frac{qr - p(1 - r)}{r(1 - r)}.$$

With $p = 0.00008$, $q = 0.99992$, $r = 0.00018$,

$$f^* \approx \frac{0.99992 \cdot 0.00018 - 0.00008 \cdot (1 - 0.00018)}{0.00018(1 - 0.00018)} \approx \frac{0.00010000}{0.00017997} \approx \boxed{0.556}.$$

So the Kelly-optimal policy limit is about 55.6% of current bankroll per contest.